

# DS Semester Examination 2023

Data Structures (DS) · Semester Examination · 2023

|            |    |      |         |
|------------|----|------|---------|
| Max. Marks | 75 | Time | 3 hours |
|------------|----|------|---------|

Answer ALL questions from Part A and any FIVE from Part B.

## Part A (25 marks)

1. Compare Asymptotic Notation with Polynomial Representation. When would you prefer one over the other? (5 marks)
2. Discuss the role of Stack ADT in Stacks & Queues. (5 marks)
3. What is meant by Binary Tree Traversals? Discuss its advantages and limitations. (5 marks)
4. What is meant by Graph Representations? Discuss its advantages and limitations. (5 marks)
5. Illustrate Binary Search with a worked example. (5 marks)

## Part B

1. Compare Asymptotic Notation with Singly Linked Lists. When would you prefer one over the other? (Unit 1)
2. Write short notes on Arrays. (Unit 1)
3. Define Stack ADT. Derive or justify the key properties involved. (Unit 2)
4. Illustrate Infix to Postfix Conversion with a worked example. (Unit 2)
5. What is meant by Binary Tree Traversals? Discuss its advantages and limitations. (Unit 3)
6. Define Binary Search Trees. Derive or justify the key properties involved. (Unit 3)
7. Write short notes on Graph Representations. (Unit 4)
8. What is meant by Breadth-First Search? Discuss its advantages and limitations. (Unit 4)
9. Discuss the role of Binary Search in Sorting, Searching & Hashing. (Unit 5)
10. What is meant by Quick Sort? Discuss its advantages and limitations. (Unit 5)